METRIC

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SUPERSEDING
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PERFORMANCE SPECIFICATION

LUBRICATING OIL, FOOD GRADE, FOOD PROCESSING EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lubricating oils for use in food processing equipment in applications where there may be incidental contact with food.

1.2 Classification. Lubricating oils for use in food processing equipment are of the following types, as specified (see 6.2):

a. Type I – Lubricating oil, general purpose

b. Type II – Lubricating oil, gears

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) NAVSEA PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D665 - Standard Test Method for Rust-Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water

ASTM D2266 - Standard Test Method for Wear Preventive Characteristics of Lubricating Grease (Four-Ball Method)

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D5949 - Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)

(Copies of these documents are available online at www.astm.org.)

NATIONAL SANITATION FOUNDATION (NSF)

Nonfood Compounds: List of Product Category Codes

(Copies of this document are available online at https://www.nsf.org.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The lubricating oil shall be a stable homogeneous blend of virgin petroleum lubricating oil stocks plus additive agents, if necessary, to meet the requirements of this specification.

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3.2 Chemical and physical requirements. The lubricating oil shall conform to the chemical and physical requirements specified in table I.

TABLE I. Chemical and physical requirements.

Characteristics	Type I Limits	Type II Limits
Viscosity, at 40 °C, mm²/s,	28-76	195-260
Viscosity, at 100 °C, mm²/s,	5.4-8.8	15.5-22.7
Pour point, °C, max	-10	-12
Flash point, °C, min	204	204
Corrosion protection	Pass	Pass
Wear characteristics, mm diameter, max	0.55	0.55
NSF rating	H-1	H-1

3.3 National Sanitation Foundation (NSF) rating. The lubricating oil shall be listed in NSF under nonfood compounds and rated H-1, lubricants approved for incidental contact with food.

3.4 Prohibited materials. The lubricating oil shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.5 Toxicity. When evaluated in accordance with 4.7, the lubricating oil shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7 and 6.4).

4. VERIFICATION

4.1 Conformance inspection. The lubricating oil shall be inspected and tested to verify compliance with the requirements specified herein. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.7.

4.2 Lot.

4.2.1 Bulk lot. A bulk lot is a homogenous mixture of one type of oil offered for acceptance in a single, isolated container; or manufactured in a single plan run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.2.2 Packaged lot. A packaged lot shall consist of unit containers of identical size and type, offered for acceptance, and filled with a homogenous mixture of one type of oil from a single, isolated container; or filled with a homogenous mixture of one type of oil manufactured in a single plan run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling. Samples shall be taken from bulk or packaged lots for tests in accordance with ASTM D4057.

4.4 Examination of filled containers. Samples of filled containers selected from each lot in accordance with MIL-STD-1916 as specified (see 6.2) shall be examined to verify compliance with all requirements of this specification regarding fill, closure, marking, and other requirements not involving tests. Containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unacceptable markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or containing less product than stated on the container (under required fill) shall be cause for rejection of the container and if the number of defective containers in any sample exceeds the acceptable number for the appropriate sampling plan specified (see 6.2), this shall be cause for rejection of the lot represented by the sample.

4.5 Testing methods. Test methods for chemical and physical requirements shall be in accordance with table II.

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TABLE II. Chemical and physical test methods.

Characteristics	Test Method or Verification
Viscosity, at 40 °C, mm^2/s,	ASTM D445
Viscosity, at 100 °C, mm^2/s,	ASTM D445
Pour point, °C, max	ASTM D97 (R), ASTM D5949 1/
Flash point, °C, min	ASTM D92
Corrosion protection	ASTM D665
Wear characteristics, mm diameter, max	ASTM D2266
NSF rating	NSF Food Grade Authorization Program, Nonfood Compounds Category

NOTE:
1/ The referee test method is identified first, followed by the notation "(R)".

4.6 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4 and 6.2).

4.7 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.5 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the lubricating oil using data provided by the manufacturer/distributor to the NMCPHC (see 3.5 and 6.4).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The oils covered by this specification are intended for use as a lubricant in food processing equipment. Current stock of lubricating oil may be used until depleted; future requisitions for lubricating oil should refer to the current version of this specification. Restrictions on composition, toxicity and hazardous materials covered by this specification cause the grease to be military unique.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of oil (see 1.2).

c. Requirement for prohibited materials verification (see 3.4 and 4.6).

d. Requirements for an HHA (see 3.5 and 4.7).

e. Sampling plan for the examination of filled containers and acceptable number of defective containers (see 4.4).

f. Packaging requirements (see 5.1).

g. When a Safety Data Sheet is required (see 6.3).

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6.3 Safety data sheet. When specified (see 6.2), the contracting activity should be provided a Safety Data Sheet (SDS) at the time of contract award. The SDS is form OSHA-20 and found as part of FED-STD-313. The SDS should be included with each shipment of the material covered by this specification.

6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.5 Subject term (key word) listing.

Nonfood compounds

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army –CR4
Navy – SH
Air Force –68

Preparing activity:
Navy – SH
(Project 9150-2020-009)

Review activity:
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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